

SIMATS ENGINEERING

ASSESSMENT TOOL 3

COURSE CODE: ITA 06

COURSE NAME: MACHINE LEARNING

COURSE OUTCOME COVERED

CO2: *Neural Network Representation, Perceptron Model, Multilayer Perceptron's, Backpropagation Algorithm, Deep Neural Networks, Genetic Algorithms, Hypothesis Space Search, Genetic Programming, Models of Evaluation and Learning (BL1, BL2)*

Assessment Tool Weightage: 10%

QUESTIONS:15

Total Marks:25

Annexure A

DEBUGGING & OPTIMIZATION TASK QUESTIONS

Code of Conduct:

I Pravin R (192421292) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

Annexure A

Short Answer Test

1. A perceptron model is trained on a dataset that is known to be linearly separable, yet the algorithm fails to converge even after many iterations. Analyze the possible causes of this issue, including improper learning rate selection, incorrect weight initialization, bias handling errors, or flawed implementation of the update rule. Propose debugging strategies to identify the root cause and suggest optimization techniques to ensure faster and stable convergence. (5 Marks)
2. While training a multilayer perceptron using the backpropagation algorithm, the network exhibits slow convergence and gets stuck in local minima. Examine the potential reasons such as vanishing gradients, inappropriate activation functions, poor weight initialization, or suboptimal learning rates. Describe step-by-step debugging approaches to trace the issue and recommend optimization techniques like adaptive learning rates, normalization, or advanced optimizers to improve performance. (5 Marks)
3. A genetic algorithm designed for hypothesis space search is producing inconsistent and suboptimal solutions across multiple runs. Investigate possible issues such as improper encoding of chromosomes, incorrect fitness function design, premature convergence, or lack of diversity in the population. Explain how to debug these problems systematically and suggest optimization strategies like mutation tuning, selection pressure adjustment, and elitism to enhance solution quality and convergence speed. (5 Marks)
4. A deep neural network combined with genetic programming is used for solving a complex prediction problem, but it suffers from overfitting and high computational cost. Identify debugging steps to analyze issues related to model complexity, over-parameterization, and insufficient training data. Propose optimization techniques such as regularization, dropout, pruning, and efficient evaluation models to balance accuracy and computational efficiency while improving generalization capability. (5 Marks)
5. A decision tree model trained on a large dataset shows high accuracy on training data but poor performance on unseen data. Analyze possible causes such as overfitting, deep tree structure, or noisy data. Propose debugging steps and optimization methods like pruning, cross-validation, and feature selection to improve generalization. (5 Marks)
6. A convolutional neural network (CNN) used for image classification produces unstable results across training epochs. Examine potential issues such as improper data augmentation, vanishing gradients, or batch normalization errors. Describe debugging strategies and suggest optimization techniques to stabilize training and improve accuracy. (5 Marks)

7. A reinforcement learning agent fails to learn an optimal policy in a simulated environment. Investigate causes such as poor reward design, insufficient exploration, or incorrect state representation. Explain debugging methods and recommend improvements like reward shaping, exploration strategies, and tuning hyperparameters. (5 Marks)

8. A clustering algorithm like K-means produces inconsistent cluster assignments for similar datasets. Analyze possible reasons such as poor initialization, incorrect number of clusters, or sensitivity to outliers. Propose debugging approaches and optimization strategies like K-means++, normalization, and silhouette analysis. (5 Marks)

9. A support vector machine (SVM) model underperforms on a nonlinear dataset. Identify issues such as incorrect kernel choice, improper parameter tuning, or scaling problems. Suggest debugging techniques and optimization methods including kernel selection, hyperparameter tuning, and feature scaling. (5 Marks)

10. A natural language processing (NLP) model using recurrent neural networks (RNNs) suffers from poor long-term dependency learning. Examine causes such as vanishing gradients and limited memory capacity. Describe debugging steps and propose solutions like LSTM/GRU units, attention mechanisms, and improved training strategies. (5 Marks)

11. A support vector machine (SVM) model trained for classification is producing very low accuracy on both training and test datasets. Analyze possible causes such as improper kernel selection, incorrect hyperparameter tuning (C and gamma), or poor feature scaling. Describe systematic debugging steps to identify the issue and suggest optimization techniques like kernel tuning, normalization, and grid search to improve model performance. (5 Marks)

12. A K-Means clustering algorithm is applied to a dataset, but the clusters formed are highly imbalanced and do not reflect the actual data distribution. Investigate possible issues such as incorrect choice of K, poor initialization of centroids, or presence of outliers. Explain debugging approaches and suggest improvements like the Elbow method, K-Means++, and outlier removal techniques. (5 Marks)

13. A recurrent neural network (RNN) designed for time-series prediction shows inconsistent outputs and fails to capture long-term dependencies. Analyze possible causes such as vanishing gradients, insufficient sequence length, or improper architecture design. Describe debugging steps and propose optimization techniques such as using LSTM/GRU units, gradient clipping, and sequence normalization. (5 Marks)

14. A Naïve Bayes classifier used for text classification performs poorly when applied to real-world data containing noisy and irrelevant features. Examine possible issues such as violation of feature independence assumption, improper preprocessing, or imbalanced dataset. Explain debugging strategies and suggest improvements like feature selection, smoothing techniques, and data balancing methods. (5 Marks)

15. A reinforcement learning agent trained to solve a navigation problem fails to learn an optimal policy and shows erratic behavior. Analyze possible causes such as sparse rewards, improper exploration-exploitation balance, or incorrect reward function design. Describe debugging approaches and recommend optimization strategies like reward shaping, epsilon decay tuning, and policy optimization methods to improve learning performance. (5 Marks)

RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 – Exemplary (5 Marks)	Level 3 – Proficient (4 Marks)	Level 2 – Developing (2–3 Marks)	Level 1 – Beginning (0–1 Mark)
Conceptual Understanding	30	Demonstrates clear and complete understanding of the concept	Good understanding with minor gaps	Basic understanding with some misconceptions	Poor or incorrect understanding
Accuracy of Answer	20	Completely correct answer with no errors	Mostly correct with minor errors	Partially correct with noticeable errors	Incorrect or irrelevant answer
Application of Knowledge	30	Correctly applies concepts to solve the question/problem	Applies concepts with minor mistakes	Limited or partially correct application	No or incorrect application
Completeness of Response	10	Covers all required parts of the question	Covers most parts with minor omissions	Covers only some parts	Incomplete or missing response
Clarity & Presentation	10	Clear, concise, and well-organized answer	Understandable with minor issues	Somewhat unclear or poorly structured	Difficult to understand

Q9 – SVM (Loan Approval)

Dataset

Customer_ID	Age	Annual_Income	Credit_Score	Loan_Approved
101	22	250000	620	0
102	25	300000	650	0
103	28	450000	700	1
104	30	500000	720	1
105	35	600000	750	1
106	40	800000	780	1
107	45	350000	640	0
108	50	900000	800	1
109	27	280000	630	0
110	32	550000	730	1
111	38	700000	760	1
112	29	480000	710	1
113	41	850000	790	1
114	36	620000	740	1
115	24	270000	625	0

Code and Output

```
import pandas as pd

data = {
    "Customer_ID": [101,102,103,104,105,106,107,108,109,110,111,112,113,114,115],
    "Age": [22,25,28,30,35,40,45,50,27,32,38,29,41,36,24],
    "Annual_Income": [250000,300000,450000,500000,600000,800000,350000,900000,280000,550000,700000,480000,850000,620000,270000],
    "Credit_Score": [620,650,700,720,750,780,640,800,630,730,760,710,790,740,625],
    "Loan_Approved": [0,0,1,1,1,1,0,1,0,1,1,1,1,1,0]
}

df = pd.DataFrame(data)

print(df)

df.to_csv("loan_approval_dataset.csv", index=False)
```

```
***   Customer_ID  Age  Annual_Income  Credit_Score  Loan_Approved
0         101    22      250000         620           0
1         102    25      300000         650           0
2         103    28      450000         700           1
3         104    30      500000         720           1
4         105    35      600000         750           1
5         106    40      800000         780           1
6         107    45      350000         640           0
7         108    50      900000         800           1
8         109    27      280000         630           0
9         110    32      550000         730           1
10        111    38      700000         760           1
11        112    29      480000         710           1
12        113    41      850000         790           1
13        114    36      620000         740           1
14        115    24      270000         625           0
```

Q10 – NLP (Sentiment Analysis)

Dataset

Review	Sentiment
The product quality is excellent.	Positive
I am very happy with the purchase.	Positive

Worst product I have ever bought.	Negative
Delivery was delayed and disappointing.	Negative
Amazing customer support.	Positive
The phone battery drains quickly.	Negative
Highly recommend this product.	Positive
Completely waste of money.	Negative
The laptop performance is outstanding.	Positive
Very poor build quality.	Negative
Fast delivery and excellent packaging.	Positive
Customer service was unhelpful.	Negative
Battery backup is amazing.	Positive
Screen quality is poor.	Negative
Value for money.	Positive

Code and Output

```
import pandas as pd
import os

data = {
    "Review": [
        "The product quality is excellent.",
        "I am very happy with the purchase.",
        "Worst product I have ever bought.",
        "Delivery was delayed and disappointing.",
        "Amazing customer support.",
        "The phone battery drains quickly.",
        "Highly recommend this product.",
        "Completely waste of money.",
        "The laptop performance is outstanding.",
        "Very poor build quality.",
        "Fast delivery and excellent packaging.",
        "Customer service was unhelpful.",
        "Battery backup is amazing.",
        "Screen quality is poor.",
        "Value for money."
    ],
    "Sentiment": [
        "Positive",
        "Positive",
        "Negative",
        "Negative",
        "Positive",
        "Negative",
        "Positive",
        "Negative",
        "Positive",
        "Negative",
        "Positive",
        "Negative",
        "Positive",
        "Negative",
        "Positive"
    ]
}

df = pd.DataFrame(data)
print("=== Sentiment Dataset ===")
print(df)

file_name = "sentiment_dataset.csv"
df.to_csv(file_name, index=False)

print("\nFile saved:", os.path.exists(file_name))

df_check = pd.read_csv(file_name)
print("\n=== Reloaded CSV Preview ===")
print(df_check.head())

print("\n=== Quick Stats ===")
print("Total Reviews:", len(df))
print("Positive Reviews:", (df['Sentiment'] == "Positive").sum())
print("Negative Reviews:", (df['Sentiment'] == "Negative").sum())
print("Positive Sentiment %:", (df['Sentiment'] == "Positive").mean() * 100, "%")

***
=== Sentiment Dataset ===
      Review Sentiment
0  The product quality is excellent. Positive
1  I am very happy with the purchase. Positive
2  Worst product I have ever bought. Negative
3  Delivery was delayed and disappointing. Negative
4  Amazing customer support. Positive
5  The phone battery drains quickly. Negative
6  Highly recommend this product. Positive
7  Completely waste of money. Negative
8  The laptop performance is outstanding. Positive
9  Very poor build quality. Negative
10 Fast delivery and excellent packaging. Positive
11 Customer service was unhelpful. Negative
12 Battery backup is amazing. Positive
13 Screen quality is poor. Negative
14 Value for money. Positive

File saved: True

=== Reloaded CSV Preview ===
      Review Sentiment
0  The product quality is excellent. Positive
1  I am very happy with the purchase. Positive
2  Worst product I have ever bought. Negative
3  Delivery was delayed and disappointing. Negative
4  Amazing customer support. Positive

=== Quick Stats ===
Total Reviews: 15
Positive Reviews: 8
Negative Reviews: 7
Positive Sentiment %: 53.333333333333336 %
```


Q11 – SVM (Student Placement)

Student_ID	CGPA
1	6.5
2	7.0
3	7.5
4	8.0
5	8.5
6	9.0
7	6.8
8	7.8
9	8.2
10	6.9
11	8.8
12	7.2
13	7.9
14	8.4
15	6.7

Code and Output

```
import pandas as pd

data = {
    "Student_ID": [1,2,3,4,5,6,7,8,9,10,11,12,13,14,15],
    "CGPA": [6.5,7.0,7.5,8.0,8.5,9.0,6.8,7.8,8.2,6.9,8.8,7.2,7.9,8.4,6.7],
    "Aptitude_Score": [60,68,72,80,85,92,62,78,84,64,90,70,81,87,63],
    "Interview_Score": [58,65,75,82,88,95,60,80,86,62,93,71,83,89,61],
    "Placed": [0,0,1,1,1,1,0,1,1,0,1,0,1,1,0]
}

df = pd.DataFrame(data)

print(df)

df.to_csv("student_placement_dataset.csv", index=False)
```

```
***
   Student_ID  CGPA  Aptitude_Score  Interview_Score  Placed
0           1   6.5             60             58         0
1           2   7.0             68             65         0
2           3   7.5             72             75         1
3           4   8.0             80             82         1
4           5   8.5             85             88         1
5           6   9.0             92             95         1
6           7   6.8             62             60         0
7           8   7.8             78             80         1
8           9   8.2             84             86         1
9          10   6.9             64             62         0
10          11   8.8             90             93         1
11          12   7.2             70             71         0
12          13   7.9             81             83         1
13          14   8.4             87             89         1
14          15   6.7             63             61         0
```

Q12 – K-Means (Mall Customer Segmentation)

Customer_ID	Age
1	19
2	21
3	23
4	25
5	28
6	31

7	35
8	38
9	40
10	42
11	45
12	48
13	50
14	55
15	60

Code and Output

```

import pandas as pd

data = {
    "Customer_ID": [1,2,3,4,5,6,7,8,9,10,11,12,13,14,15],
    "Age": [19,21,23,25,28,31,35,38,40,42,45,48,50,55,60],
    "Annual_Income": [15000,18000,22000,25000,30000,40000,50000,55000,65000,70000,80000,85000,90000,95000,100000],
    "Spending_Score": [39,42,35,45,50,55,60,65,70,75,82,88,91,94,97]
}

df = pd.DataFrame(data)

print(df)

df.to_csv("customer_segmentation_dataset.csv", index=False)

```

```

***
   Customer_ID  Age  Annual_Income  Spending_Score
0             1   19         15000             39
1             2   21         18000             42
2             3   23         22000             35
3             4   25         25000             45
4             5   28         30000             50
5             6   31         40000             55
6             7   35         50000             60
7             8   38         55000             65
8             9   40         65000             70
9            10   42         70000             75
10           11   45         80000             82
11           12   48         85000             88
12           13   50         90000             91
13           14   55         95000             94
14           15   60        100000             97

```